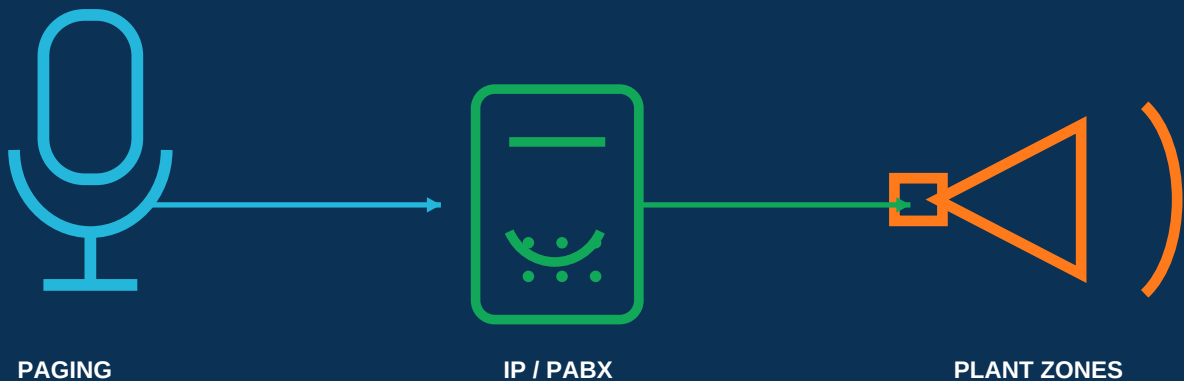


PA Solution

Design, Deployment & Handover

Seven-zone IP paging, flameproof telephony and PABX integration.



CONCEPTUAL SIGNAL FLOW - DETAILED SKETCH INSIDE

27

INDOOR + OUTDOOR
SPEAKERS

12

FLAMEPROOF TELEPHONES

7

PAGING ZONES

2.745 km

ARMOURED CAT6

Executive Success Story

THE TRANSFORMATION

One IP communication layer for office and process plant

D-TECH designed, installed and commissioned an IP-based PA and paging solution integrating indoor announcements, outdoor high-output coverage, flameproof field telephones, PoE networking, zone control and the existing GENX PABX environment.

9 + 18

INDOOR + HORN SPEAKERS

12

FLAMEPROOF FIELD PHONES

4

MANAGED POE SWITCHES

7

PROGRAMMED PA ZONES

BUSINESS NEED

Clear, targeted announcements across mixed-risk areas.

The site required intelligible voice communication across administration, utilities, warehouse, process zones and MPP areas. The design had to combine safe outdoor devices, zone-wise selection, central paging and compatibility with the existing telephony platform.

01

Designed

Coverage, zones, interfaces and field-device architecture

02

Deployed

Rack, PoE, Cat6, speakers, phones and paging consoles

03

Integrated

VoIP gateway, SIP users and existing GENX PABX

04

Accepted

Live tests, documentation and signed customer handover

Approx. documented value USD 0.018 million equivalent (rounded presentation value, including applicable taxes).

Requirements & Design Basis

THE CHALLENGE

Different spaces needed different acoustic and safety responses

A single speaker type could not meet the requirement. The solution had to address quiet indoor rooms, high-noise outdoor spaces, and selective paging.

01

Indoor intelligibility

Wall-mount speakers for admin, security, OHC, utility and control spaces.

02

Outdoor projection

30 W horn speakers for process, utility, warehouse and MPP coverage.

03

Hazardous-area calling

IP65 flameproof field telephones suitable for IIA/IIB process environments.

04

Selective paging

Seven programmed zones plus an all-area broadcast path.

05

PABX integration

Connect paging and field communication with the existing GENX telephony system.

06

IP transport

PoE switching and armoured Cat6 for power, voice and management.

07

Operational control

Admin and security paging consoles for authorised announcements.

08

Evidence & handover

IP lists, serial mapping, extension schedules, test records and acceptance.

DESIGN PRINCIPLE

Place the right transducer in the right acoustic zone, then make every endpoint addressable and testable.

SIGNAL FLOW

From authorised paging source to targeted plant zones

The sketch shows how paging consoles and flameproof telephones use the existing PABX and VoIP gateway, then traverse the managed PoE network to individually addressed indoor and outdoor speakers.



Conceptual sketch derived from the commissioned architecture; not a construction drawing.

Paging Zone Coverage Plan

ZONE LOGIC

Seven callable zones plus an all-area broadcast

The programmed paging map lets authorised users reach a specific operational cluster or issue a controlled site-wide announcement when required.



Solution Scope & Bill of Capability

WHAT WAS DELIVERED

Integrated voice, paging, network and field hardware

The awarded scope combined specialist communication devices, standard network components, protected field installation and software licensing into one commissioned solution.

SOLUTION LAYER	SCALE	DELIVERED CAPABILITY
Indoor audio	9 speakers	15 W full-range wall-mount network speakers
Outdoor audio	18 speakers	30 W high-power IP network horn speakers
Hazardous-area voice	12 phones	IP65 flameproof paging telephones with external sounder
Paging control	3 terminals	IP paging/intercom terminals; two active consoles documented
PoE network	4 switches	Two 24-port and two 8-port Gigabit PoE switches
Structured cabling	2,745 m	Armoured Cat6, patching, I/Os, connectors and tagging
Voice integration	1 gateway	16-port VoIP-FXO-FXS gateway with SIP/PABX integration
Software	Licences	IP subscribers and 16 simultaneous transcoding channels
Field mounting	18 sets	Fabricated stands and certified IP66 junction boxes
Rack & service	Turnkey	9U cabinet, installation, testing, documentation and handover
PROJECT REFERENCE PO D/FA/26-27/45097 dated 29 May 2026; completion recorded 3 September 2026.		

FROM DESIGN TO LIVE ANNOUNCEMENT

A disciplined eight-step implementation model

The project combined voice engineering, IP configuration and physical installation. Each stage created evidence required for

01

Survey & acoustic zoning

Validate noise, mounting positions, device type, cable routes and safe access.

02

Low-level design

Define zones, IP plan, extensions, speaker mapping, rack and switch ports.

03

Pathway & cabling

Deploy armoured Cat6, protect routes, maintain separation and tag cables.

04

Rack & PoE layer

Install 9U rack, patching, cable manager, PoE switches and uplinks.

05

Field-device mounting

Install wall speakers, horn stands, IP66 boxes and flameproof phones.

06

PABX / SIP integration

Configure gateway, register devices, licences and paging paths.

07

Testing & tuning

Check zone selection, clarity, coverage, endpoint calls and network health.

08

Handover

Submit IP/serial lists, zone map, extensions, tests and signed acceptance.

QUALITY GATE

A zone is complete only after the device, network path, paging code and audio coverage all pass together.

TEST PLAN

Technical validation from console to speaker and field telephone

Commissioning proved signalling, addressing, zoning, audio intelligibility and telephone operation before the customer

01

SIP registration

Verify each IP speaker and console is registered and reachable.

02

Zone selection

Confirm codes 1-6 address the intended area and code 0 reaches all zones.

03

Indoor clarity

Validate speech intelligibility and appropriate level on 15 W speakers.

04

Outdoor coverage

Confirm even projection and usable speech from 30 W horn speakers.

05

FLP telephone calls

Test all 12 extensions, audio path, keypad and external sounder.

06

Network health

Check PoE stability, link status, IP addressing and device response.

07

PABX integration

Validate gateway, extension plan, paging calls and simultaneous paths.

08

Documentation

Reconcile IP, MAC, serial, zone, dial-code and extension records.

ACCEPTANCE EVIDENCE

The signed report records joint inspection, satisfactory completion and operation as intended at the time of inspection. A client-signed acceptance excerpt is reproduced on the final page.

Operating Model & Use Cases

DAY-TO-DAY VALUE

A communication platform for operations and emergency response

The solution supports routine announcements, targeted coordination and urgent communication while keeping control with

01

Shift communication

Broadcast reporting times, shift changes and coordinated instructions.

02

Process alerts

Reach selected operating areas without disturbing the complete site.

03

Safety announcements

Issue clear instructions during drills, incidents or restricted activity.

04

Warehouse coordination

Communicate with loading, dispatch and storage teams.

05

Utility response

Page DCS, MCC, boiler, ETP and utility groups using dedicated zones.

06

Field calling

Use flameproof telephones where standard handsets are unsuitable.

07

Security control

Authorised security staff can make targeted or all-area announcements.

08

Admin paging

Management can communicate with office and plant groups through one platform.

CONTROLLED COMMUNICATION

Zone codes make messages relevant; all-area paging remains available for authorised site-wide use.

OUTCOMES

A safer, faster and more manageable communication layer

The design creates value across safety, production coordination, maintainability and future expansion - not simply higher

01

Coverage fit

Indoor and outdoor speaker types are matched to their acoustic environments.

02

Targeted messaging

Seven zones reduce unnecessary announcements and improve relevance.

03

Hazardous-area access

Certified field telephones extend communication into process areas.

04

Central control

Admin and security operate paging from authorised IP consoles.

05

Existing-system value

PABX integration protects prior investment and simplifies adoption.

06

Simplified cabling

PoE and IP transport consolidate power, voice and management paths.

07

Serviceability

Addressable devices, labels and mapping accelerate troubleshooting.

08

Expansion readiness

IP architecture supports additional zones, speakers and consoles.

SUCCESS INDICATOR

The commissioned system passed joint inspection and was accepted as operating as intended.

Evidence & Repeatable Blueprint

WHY D-TECH

One accountable partner across PA, voice and network layers

This project demonstrates D-TECH's ability to translate a communication requirement into an integrated design, execute safely in mixed environments, integrate with existing systems and close with documented customer acceptance.

D-TECH CAPABILITY

Design | Supply | Integration | Testing | Handover

Acoustic zoning | IP paging | Hazardous-area telephony | PABX and SIP integration | PoE networks | Armoured cabling | Documentation and support

PROJECT EVIDENCE

ITEM	REFERENCE	BASIS
Award	PO D/FA/26-27/45097	29 May 2026
Project	IP PA + FLP extension	Krystal
Completion	Installation report	03 Sep 2026
Acceptance	Joint inspection	Acknowledged

SIGNED INSTALLATION ACCEPTANCE

Acknowledged & Accepted
For Krystal Ingredients Private Limited
Authorized Signatory
Name: Mr. Chirag Patel
Signature: _____
Date: 04/09/2026

Remarks: The installed systems have been jointly inspected and verified and the work has been inspected and found to be satisfactory and operating as intended at the time of inspection.

Prepared By
For M/s. D-Tech Solution Integrators Private Limited
Authorized Signatory
Name: Bhavin Parmar
Signature: _____
Date: 04/09/2026

Customer accepted - 04 September 2026

CLIENT-SAFE DISCLOSURE

Signed acceptance is shown as documentary evidence. Contact addresses, tax identifiers, IP/MAC details and unit prices are excluded. USD value is a rounded presentation equivalent.

Clear voice. Correct zone. Confident action.